

Memory Match — Exercise 2

COMP5450 Mobile Programming • Liang Mingchen

Overview

Flutter/Dart memory matching game with a 4×5 card grid (10 animal pairs). Players flip two cards per turn, match pairs, and restart anytime.

Exercise requirements

- Flutter/Dart source code
- README.pdf with configuration, run steps, screenshots, and structure
- Full project suitable for GitHub upload
- ZIP package for D2L (Dart files + README.pdf)

Configure and run

- Install Flutter SDK and run flutter doctor.
- `cd exercises/exercise-2`
- `flutter pub get`
- `flutter run` (choose Android emulator, iOS simulator, or a physical device).
- Optional: `flutter run -d chrome` for web preview.

Project structure

`lib/main.dart` - app entry

`lib/game/memory_game.dart` - shuffle, matching, moves

`lib/screens/memory_match_screen.dart` - UI

`lib/widgets/memory_card_tile.dart` - card widget

`lib/models/memory_card_data.dart` - card model

`test/widget_test.dart` - widget tests

Checks

`flutter analyze`, `flutter test`, and `flutter build web` were run successfully during development.

Screenshots

Home board (face-down cards):



Gameplay (flipped cards):



Memory Match

1

Find all matching pairs! 🐾

Matched: 0 / 10



🔄 Restart